

Profiling Harness

Code architecture, explained end to end

2349 lines / 7 Python modules / 4 profilers / 111 checks

This document walks the harness from the command line down to the process that actually runs your workload. It follows execution order rather than file order, so each piece arrives when you need it.

Section	What it covers
1 - 2	The shape of the system, and the file tree
3 - 4	Entry point, dispatch, and discovery
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How to read this

Every claim about behaviour in this document was verified by running the code, not by reading it. Where a number appears (a race that hits 1 run in 30, a 106 MB trace, a 10-second kill), it was measured. Section 17 lists the places where the implementation deliberately departs from the specification, and why.

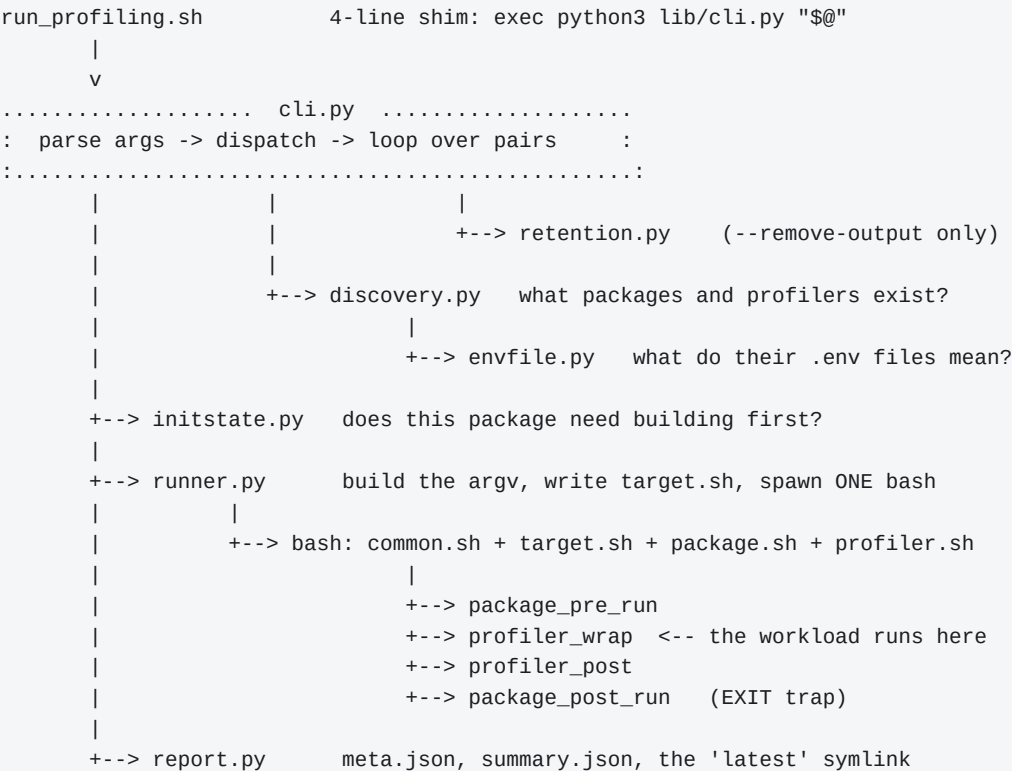
SECTION 1

The shape of the system

A pipeline, not a framework

The harness runs a project's workloads (**packages**) under measurement tools (**profilers**), writes timestamped artifacts, and reports a single exit code. Both packages and profilers are discovered from the filesystem, so adding either means creating a directory with two small files - never editing the core.

There is no plugin registry, no dependency injection, no class hierarchy. Each module owns one noun and hands its result to the next:



Data flows one way. discovery produces immutable value objects; runner consumes them and produces a result record; report serialises it. Nothing writes back up the chain. That is why the modules can be read in isolation.

The Python/bash split

Orchestration lives in Python; command composition lives in bash. The dividing line is deliberate and it is the single most important design decision in the tool.

Python does	Bash does
Argument parsing, discovery, environment layering, retention, exit-code aggregation, JSON output, process supervision	Building the actual command line, per-package setup, per-profiler invocation
Painful in bash: nested data, JSON, timeouts, signal handling	Natural in bash: quoting, arrays, sourcing config, calling tools

The consequence for a user: **anyone adding a package writes only shell**. There is no Python plugin API to learn. Python 3 was already a hard requirement because three of the four shipped profilers are Python tools, so it costs nothing.

The core imports only the standard library - verified, not assumed: argparse, dataclasses, hashlib, json, os, pathlib, platform, re, secrets, shlex, shutil, signal, subprocess, sys, tempfile, threading, time, typing. Zero third-party packages.

SECTION 2

The file tree

What each file owns

profiling/		
run_profiling.sh		the entry point (a shim)
install.sh		discovery-driven dependency installer
config.env		all paths and global defaults
README.md		user-facing documentation
.gitignore		output/ and .state/
lib/		the core - the only Python in the tree
cli.py	701	argument parsing, dispatch, the main loop
runner.py	613	the target contract, bash harnesses, process control
discovery.py	244	enumerate packages/profilers, typed access to .env
report.py	228	meta.json, summary.json, tables and JSON listings
initstate.py	207	init fingerprinting and stamps
retention.py	175	--remove-output planning and execution
envfile.py	172	source .env layers, apply precedence
common.sh		helpers sourced into every leaf hook
packages/<name>/		a workload to measure
.env		declares it
package.sh		optional hooks
profilers/<name>/		a way to measure
.env		declares it
profiler.sh		required profiler_wrap, optional post/dry_run
output/		run artifacts (gitignored)
.state/		package init stamps (gitignored)

The discovery rule

A directory under packages/ or profilers/ is a valid entry **if and only if** it contains a .env file and its name does not start with an underscore. Names must match `[a-z0-9][a-z0-9._-]*`; anything else is rejected with a clear error rather than silently ignored.

The underscore prefix is what keeps `_template/` out of the listings while still letting you `cp -r` it. A directory without a .env is ignored entirely, so editor droppings and stray folders cause no trouble.

Why state and output are separate trees

.state/ holds the answer to "has this package been built?". output/ holds run artifacts. They are separate directories, and the init log is written to .state/ rather than into a run directory, for one specific reason: **pruning artifacts must never trigger a rebuild**. If the build stamp lived under output/, then --remove-output would silently cost you a full rebuild on the next run.

It also means you can mount .state/ as a Docker volume to carry "already built" across containers, while letting output/ stay ephemeral.

SECTION 3

Entry point and dispatch

```
run_profiling.sh -> cli.py main()
```

The shim

```
#!/usr/bin/env bash
set -euo pipefail
exec python3 "$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)/lib/cli.py" "$@"
```

Four lines, and the `exec` matters. It *replaces* the shell process with Python rather than forking, so the process the user sees is the process doing the work. A `SIGTERM` sent to `run_profiling.sh` lands directly on the Python interpreter - which, as Section 10 explains, is what makes clean cancellation possible at all.

main()

`cli.py:657`. Four steps, in order:

#	Step	Why here
1	<code>parser.parse_args()</code>	<code>argparse</code> exits 2 on a bad flag, which is already the code the spec wants
2	<code>_install_signal_handlers()</code>	Before anything can spawn a child, so no window exists where a signal orphans work
3	<code>load_global_env()</code>	Sources <code>config.env</code> alone. Even <code>--remove-output</code> needs <code>PROFILING_OUTPUT_DIR</code>
4	<code>dispatch</code>	A flat if-chain, because three of four commands are terminal

Terminal actions

`--list-packages`, `--list-profilers` and `--remove-output` each print and return. They never combine with a `run`, and combining two of them is a usage error. This is why `dispatch` is a plain if-chain rather than a command table: there are four commands, three of which are one-liners, and the fourth is the entire rest of the program.

```
if args.list_packages:      return cmd_list_packages(env, args.json)
if args.list_profilers:     return cmd_list_profilers(env, args.json)
if args.remove_output is not None: return cmd_remove_output(args, env)

if args.keep is not None:   raise UsageError("--keep only with --remove-output")
if args.json:              raise UsageError("--json only with --list-*")
return cmd_run(args, env)
```

The two guards before `cmd_run` catch flags that parse fine but mean nothing in context. Silently ignoring `--keep` 5 on a `run` would be worse than refusing it - the user clearly intended something.

The selector grammar

`--package` and `--profiler` both accept four spellings, handled once in `discovery.resolve_selection`:

```
--profiler time           one name
--profiler time,py-spy    comma-separated
--profiler time --profiler py-spy repeated flag (argparse action="append")
--profiler all            the literal 'all'
```

Mixing `all` with explicit names is an error, not a silent union - `--profiler all,time` almost certainly means the user misunderstands something.

--profiler is mandatory

Omitting it while `--package` is present exits 2 with a message pointing at `--profiler all`. There is no implicit default sweep. Profiling every workload with every tool is expensive, and a harness that does it because you forgot a flag is a harness that burns CI minutes for no reason.

SECTION 4

Discovery

discovery.py - turning directories into typed objects

discovery.py does three things: scan the filesystem, load a .env stack into an environment dict, and wrap that dict in an object with typed accessors.

Scanning

```
def _scan(root, label, script_name) -> Dict[str, Entry]:
    for child in sorted(root.iterdir()):
        if not child.is_dir() or child.name.startswith(("_", ".")):
            continue # _template and hidden dirs
        if not (child / ".env").is_file():
            continue # not an entry at all
        if not NAME_RE.match(child.name):
            raise DiscoveryError(...) # a real mistake - say so
        found[child.name] = Entry(...)
```

sorted() gives deterministic order for free. The distinction between the two continue branches and the raise is deliberate: a directory with no .env is not an entry and is none of our business, but a directory that clearly *is* an entry and has an unusable name is a mistake worth stopping for.

Typed access

Package and Profiler are dataclasses wrapping the merged environment. Their properties are where defaults and parsing live, so no other module has to remember them:

```
@property
def args(self) -> List[str]:
    return shlex.split(self.env.get("PACKAGE_ARGS", "")) # not .split()

@property
def timeout(self) -> Optional[float]:
    raw = (self.env.get("PACKAGE_TIMEOUT") or "").strip()
    if not raw: return None
    value = float(raw) # raises DiscoveryError on garbage
    return None if value <= 0 else value # 0 means "no timeout"

def supports(self, kind: str) -> bool:
    return not self.kinds or kind in self.kinds # empty = no restriction
```

shlex.split rather than .split() is what makes PACKAGE_ARGS="--input 'my file.json'" work. This was tested directly against arguments containing spaces, single and double quotes, globs and dollar signs.

Selection and ordering

resolve_selection expands selectors into an ordered, de-duplicated list. Ordering follows one rule, applied independently to packages and profilers:

Given	Order
Explicit names	As typed on the command line
all	Alphabetical

So --profiler viztracer, time runs viztracer first, while --profiler all runs a package's list alphabetically no matter how it is written in PACKAGE_PROFILERS. The listing output is sorted the same way, so a CI matrix built

from `--list-packages --json` matches what the harness will actually do.

SECTION 5

Environment layering

envfile.py - the first of two central ideas

Every run builds its environment by sourcing four layers in order. Later layers win:

- | | |
|---------------------------------|-------------------------------------|
| 1. config.env | global paths and defaults |
| 2. profilers/<profiler>/ .env | that profiler's knobs |
| 3. packages/<package>/ .env | the package, ON TOP of the profiler |
| 4. the real process environment | whatever the caller exported |

Layer 3 above layer 2 is the whole point

It is what lets a package tune a profiler *for itself*. packages/my-tool/ .env can set LINE_PROFILER_TARGETS, PYPY_NATIVE=1 because it has C extensions, or a shallower VIZTRACER_MAX_DEPTH because its call graph is deep - all without touching the profiler. This is also why the package is loaded twice per pair, as Section 7 explains.

Why bash does the sourcing

The .env files are **sourced by bash, not parsed by Python**. The implementation runs one bash process and reads back its environment:

```
bash -c 'set -a; . "$1"; . "$2"; ...; env -0' _ <files...>
```

set -a auto-exports every assignment; env -0 emits NUL-separated NAME=VALUE pairs so values containing newlines survive intact. The payoff is that variable interpolation, command substitution and PATH=". . . : \$PATH" all behave exactly as an author would expect, with no parser to write and no subset of shell syntax to document.

The cost is that a .env file is code. A stray \$ gets expanded. That is a real trade, and it is documented rather than papered over.

Naming the file that failed

A non-zero exit from the sourcing subshell is a hard error. Getting the filename into that message is harder than it looks, because a *syntax* error kills the shell outright - an if ! source guard never runs. An EXIT trap catches both cases:

```
trap '
  if [ "$?" -ne 0 ]; then
    printf "__PROFILING_ENV_FAIL__%s\n" "$__profiling_current" >&2
  fi
' EXIT
```

Python looks for that marker on stderr and reports the offending path. In practice the user sees the filename twice - once from the harness and once from bash's own diagnostic with a line number - which is more useful than either alone.

The one exception to "the real environment wins"

Strict layer-4 precedence has a flaw. If a package writes PYTHONPATH="\$MY_SRC:\$PYTHONPATH", the overlay would immediately clobber it back to the inherited value, making the line a no-op. So PATH-like variables are treated as *extended* rather than assigned:

```
PATHLIKE = {PATH, PYTHONPATH, LD_LIBRARY_PATH, LD_PRELOAD,
            MANPATH, PKG_CONFIG_PATH, CPATH}
```

```
for key, value in real_env.items():
    if key in PATHLIKE and merged.get(key) != base.get(key):
        continue                # a layer changed it deliberately - keep that
    merged[key] = value          # everything else: the real environment wins
```

Every other variable follows the documented rule exactly. This exception is called out in the README because it is the one place where the four-layer model is not literally true.

SECTION 6

The target contract

runner.py - the second central idea, and the one that makes profilers pluggable

Profilers come in two incompatible shapes, and getting this right is what the whole design turns on:

Family	Example	How it takes the workload
Prefix wrapper	time, py-spy	Runs the complete command: <code>/usr/bin/time -v -- python -m tool</code>
Interpreter replacement	viztracer, kernprof	Is the interpreter: <code>viztracer -m tool</code> , never <code>viztracer python -m tool</code>

A single argv cannot serve both. So the runner exposes the workload in two forms, written to a generated file that the harness sources:

```
# <RUN_DIR>/target.sh -- generated, one shlex.quote per element
TARGET_KIND=python-module
TARGET_PYTHON=/repo/.venv/bin/python
TARGET_ARGV=(/repo/.venv/bin/python -m my_tool.cli --input data/sample.json)
TARGET_PYTHON_ARGV=(-m my_tool.cli --input data/sample.json)
```

Variable	Meaning
TARGET_ARGV	The complete plain command - exactly what you would run with no profiling. For prefix wrappers.
TARGET_PYTHON_ARGV	The same minus the interpreter. This is Python's own trailing CLI shape, so it drops straight in after an interpreter-replacing tool's flags. Unset when PACKAGE_KIND=exec.
TARGET_PYTHON	The interpreter path. Unset for exec.
TARGET_KIND	The package's PACKAGE_KIND.

Why a sourced file rather than environment variables

Bash arrays cannot be exported. Passing an argv through the environment means encoding it - and every encoding scheme eventually meets an argument containing the delimiter. Writing a file with one `shlex.quote` per element and sourcing it sidesteps the problem completely: bash re-parses its own quoting, so the array on the other side is exactly the list Python built.

This was tested with arguments containing semicolons, `$(...)`, backticks, spaces and globs. They arrive as literal argv elements, and nothing is substituted at exec time.

A second benefit: `target.sh` is left in the run directory as an artifact. To reproduce a profiled run by hand, source it and run `"${TARGET_ARGV[@]}"`.

The exec kind, and failing readably

For `PACKAGE_KIND=exec` there is no interpreter to strip, so the generated file emits `unset TARGET_PYTHON_ARGV` rather than an empty array. An empty array would expand to nothing and the profiler would complain about its own arguments, which is a baffling error to debug. Instead `common.sh` provides a guard:

```
require_python_target() {
    if ! profiling_has_python_target; then
        profiling_die \
            "profiler '$PROFILER_NAME' needs a Python target, but package" \
            "'$PACKAGE_NAME' has PACKAGE_KIND=$TARGET_KIND. Use a prefix-style" \
            "profiler (e.g. 'time'), or drop it from PACKAGE_PROFILERS."
    fi
}
```

```
}
```

Nothing shipped uses `exec` yet - only `time` declares support for it - but the path is plumbed and tested end to end, so adding `perf` or `callgrind` later needs no core change.

The escape hatch

When a command cannot be expressed as `kind + entry + args`, a package defines `package_command` in `package.sh` and populates the arrays itself. It overrides the declared entry point completely. The harness reads the final `argv` back from `bash` after the hook runs, so `meta.json` and `--dry-run` both reflect the override rather than the declaration.

SECTION 7

The main loop

cli.py:345 - cmd_run, the spine of the program

Two nested loops, packages outer and profilers inner, with an eight-step checklist inside. This is the only function in the codebase worth reading carefully.

```
for package_name in package_names:
    base_package = load_package(CONFIG_ENV, entry)          # config + package
    selected, forced = _select_profilers_for_package(...)    # per-package list

    for profiler_name in selected:
        profiler = load_profiler(CONFIG_ENV, ...)           # config + profiler
        package = load_package(CONFIG_ENV, entry,
                                profiler.entry.env_file)    # config+prof+package

        ...eight steps...
```

Why the package is loaded twice

This is the one genuinely non-obvious line in the file. The package is loaded once *alone* to read `PACKAGE_PROFILERS` - you cannot know which profilers apply until you have read the package - and then again *on top of each profiler's* `.env`. That second layering is what delivers the precedence rule from Section 5. The environment is genuinely different for every (package, profiler) pair, so it must be rebuilt for every pair.

The eight steps

#	Step	Line	On failure
1	Kind compatible?	417	Skip. Informational line, recorded as skipped, never affects the exit code
2	Required binaries present?	430	Record failed, exit code 3, continue to the next pair
3	Init needed? (once per package)	458	Skip that package's runs, exit code 4
4	Build target + allocate run id	472	Usage error, exit code 2 - a malformed package is not a run failure
5	Dry run?	479	Print the resolved command and continue - nothing is created
6	Execute	493	Status recorded as failed or timeout, exit code 1
7	Write meta.json, update 'latest'	503	-
8	Aggregate the exit code	507	<code>max()</code> - the highest category wins

Three details carry requirements that look bigger than their code

```
exit_code = max(exit_code, EXIT_RUN_FAILED)    # "report the highest code"
init_done: Dict[str, bool] = {}                # "init at most once per invocation"
if not profiler.supports(package.kind): continue # "skips never fail the build"
```

A dict, a `max()` and a `continue`. No state machine, no scheduler. The run-everything-then-aggregate behaviour falls out of never breaking the loop.

How the loop body became its own function

The eight steps above used to live inline in `cmd_run`, making it 174 lines. The defence for that was "the ordering of these steps is fragile and only visible when they are adjacent" - which is true, but does not argue for keeping them *inside the loop*. Extracting them into `_run_pair` keeps all eight adjacent and leaves `cmd_run` as a readable 71-line loop.

The real defect was not length. Six different branches each had to remember to do *two* things - append a record **and** raise the exit code - interleaved with the logic, so a branch could silently do one and forget the other. `_run_pair` returns (record, exit-category) from every path, so the two can no longer drift apart.

Profiler selection, resolved per package

The rules, in order, from `_select_profilers_for_package`:

Given	Result
<code>--profiler <name></code>	Exactly those, forced . A profiler not in the package's list still runs, warns on stderr, and is marked "forced": true in meta.json
<code>--profiler all</code>	The package's <code>PACKAGE_PROFILERS</code>
...and that is empty	<code>DEFAULT_PROFILERS</code> from <code>config.env</code>
...and that is empty too	Every discovered profiler

Because this resolves *per package*, `--package all --profiler all` is not a cartesian product. A package listing two profilers gets two runs while its neighbour listing three gets three.

SECTION 8

Package init

initstate.py - build once, and know when that stops being true

Some packages need a one-time build before they can be profiled: a virtualenv, a compile step, `uv sync --frozen`. Running that before every profiler would multiply the cost by the number of profilers; caching it forever would silently profile stale code. The fingerprint is the compromise.

```
fingerprint = sha256( packages/<pkg>/.env
                      + packages/<pkg>/package.sh
                      + contents of each PACKAGE_INIT_FINGERPRINT file )
```

The decision table

Condition	Action	Why
package.sh defines no package_init	skip	Nothing to do
--force-init	run	Explicit override
No stamp file	run	Fresh container
Fingerprint changed	run	Lockfile or entry point moved
Previous status was not ok	run	A failed build is never cached as done
Otherwise	skip	Already built, inputs unchanged

Three decisions worth explaining

A missing fingerprint file is a hard error

If `PACKAGE_INIT_FINGERPRINT` lists a path that does not exist, the harness stops with a message naming it. The tempting alternative - hash it as empty - means a typo produces a fingerprint that never changes again, so init silently never re-runs. The failure would surface weeks later as "why is CI profiling last month's code". Loud beats convenient.

The log goes to .state/, never the run directory

Covered in Section 2: if the build log lived under `output/`, pruning would cost a rebuild.

A failed init writes a stamp too

It records `"status": "failed"`. The next run sees a non-ok status and retries. Writing nothing would also work, but writing the failure keeps the log and the exit code together for whoever is debugging it.

Verified behaviour

Measured, not assumed - deleting `.state/` and then touching a fingerprinted source file produces exactly this sequence of init counts across five consecutive runs:

```
fresh .state/      -> 1 init
run again          -> 0
touch a source file -> 1
run again          -> 0
--force-init       -> 1
```

SECTION 9

The bash harness

Where Python stops and the workload begins

`runner.execute` writes `target.sh`, then spawns **one** bash process running a fixed script. One process, not several, because the hooks share shell state and because a single process is a single process group to kill.

```
. "$PROFILING_ROOT/lib/common.sh"      # helpers: run_artifact, require_bin, ...
. "$RUN_DIR/target.sh"                # TARGET_ARGV, TARGET_PYTHON_ARGV, ...
. "$PROFILING_PACKAGE_SH"             # package hooks (if the file exists)
. "$PROFILING_PROFILER_SH"            # profiler hooks (if the file exists)

declare -F package_command && package_command      # escape hatch may override
printf '%s\\0' "${TARGET_ARGV[@]}" > "$PROFILING_ARGV_FILE" # report the truth

declare -F profiler_wrap || { profiling_error "no profiler_wrap"; exit 78; }

__profiling_post_run() { declare -F package_post_run && package_post_run; }
trap __profiling_post_run EXIT                # <-- fires even on failure or kill

cd "$PACKAGE_WORKDIR" || exit 77

declare -F package_pre_run && package_pre_run || exit $?

profiler_wrap "${TARGET_ARGV[@]}"            # <-- THE WORKLOAD RUNS HERE
__profiling_status=$?

[ $__profiling_status -eq 0 ] && declare -F profiler_post && profiler_post

exit "$__profiling_status"                  # the workload's status, unmodified
```

Four properties this arrangement buys

Property	How
<code>package_post_run</code> runs even when the workload dies	It is an EXIT trap, not a line after the call. Verified for a failing workload, a failing <code>pre_run</code> , and a timeout kill
The workload's exit status survives untouched	<code>profiler_wrap</code> 's status is captured immediately and is what the script exits with. A <code>pre_run</code> that returns 42 produces <code>exit_code: 42</code> in <code>meta.json</code>
<code>profiler_post</code> only runs on success	Guarded on the captured status. Post-processing a profile that was never written produces confusing errors
Every hook is optional	<code>declare -F</code> before each call, so <code>package.sh</code> may be an empty file

Reading the argv back

The harness writes the final `TARGET_ARGV` to a NUL-separated file that Python reads afterwards. This exists so that `meta.json` records the command that *actually ran* - including any `package_command` rewrite - rather than the one Python guessed before the hooks had their say. The file is deleted once read, and dotfiles at the top of a run directory are excluded from the artifact list regardless.

--dry-run and the honest limit on exactness

"Print the exact command each profiler would execute" is not achievable for an arbitrary bash function without that function's cooperation - the command only exists once the hook has expanded its variables. Rather than approximate it, the contract adds an optional hook:


```
_pyspy_build_cmd() { cmd=(py-spy record --rate "$PYSPY_RATE" ...); }

profiler_wrap()      { local cmd; _pyspy_build_cmd; "${cmd[@]}; }
profiler_dry_run()   { local cmd; _pyspy_build_cmd; profiling_show_command "${cmd[@]}; }
```

Both paths share one argv builder, so the printed command provably cannot drift from the executed one. All four shipped profilers implement it. A third-party profiler that does not gets a clearly-labelled fallback naming the wrapper, rather than a plausible-looking guess.

--dry-run creates no directories and starts no workload. It writes `target.sh` to a temporary directory, prints the run directory it *would* have used, and exits 0 even when the run would have failed.

SECTION 10

Process control

Tee, timeout, and not leaking processes

Tee-ing output

Two daemon threads copy the child's stdout and stderr to the console and to `stdout.log / stderr.log` simultaneously, line by line with an explicit flush. You watch the workload live *and* get the capture. Verified at 50,000 lines with a matching checksum against a direct run, and with multi-byte UTF-8 output.

Timeout

The child is spawned with `start_new_session=True`, giving it its own process group so that a timeout can take the profiler *and* everything it spawned down together. On expiry the group is signalled and the run is recorded as `"status": "timeout"`.

That same isolation caused a real bug

Because the child is in its own session, a signal sent to the harness does **not** reach it. SIGINT, SIGTERM and SIGHUP each left py-spy and the workload running in the container - a cancelled CI job would leak a profiler burning CPU. All three are now handled explicitly and kill the process group on the way out.

And underneath it, a second one

The first fix still leaked. Escalation to SIGKILL was conditioned on the *direct child* still being alive - but GNU time sets SIGTERM to SIG_IGN, and an ignored disposition survives exec. So `time -- sleep 600` left a sleep that shrugged off the very signal that killed its parent bash, and SIGKILL never came.

The fix keys escalation on whether the *process group* is empty, probed with signal 0:

```
_signal_group(pgid, SIGTERM)

proc.wait(timeout=term_grace) # reap the child FIRST - see below

if _group_alive(pgid):        # anything left ignored SIGTERM
    _signal_group(pgid, SIGKILL)
    while time.monotonic() < deadline and _group_alive(pgid):
        time.sleep(0.05)
```

The `proc.wait()` before the probe is not decoration. An un-reaped zombie is *still a member of its process group*, so probing first always reported the group alive and stalled every kill for the full grace period - a 3-second timeout was taking 10 seconds of wall clock. Reaping first brings it to about 3.

The roughly 2 seconds that remain are a deliberate courtesy window: a workload that handles SIGTERM gets a chance to flush partial results before SIGKILL lands.

Verified by process-group id across SIGINT, SIGTERM, SIGHUP and the timeout path, for both a plain exec workload and py-spy's two-process attach mode. Zero survivors in every case.

SECTION 11

Output artifacts

report.py - what a run leaves behind

```
output/
  <package>/<profiler>/<run-id>/
    meta.json      what happened
    target.sh      the exact command, re-sourceable by hand
    stdout.log     tee'd
    stderr.log     tee'd
    <artifacts>    profile.svg, trace.json, out.lprof, time.txt, ...
  <package>/<profiler>/latest -> <run-id>      relative symlink
  summary.json    last invocation, overwritten
```

run-id is YYYYmmdd-HHMMSS-<4 hex>. The RUN_ID environment variable overrides it wholesale so CI can use a build number; injected values are sanitised down to [A-Za-z0-9._-], so a value like ../../etc/passwd cannot escape the output tree.

meta.json

```
{
  "package": "my-tool", "profiler": "py-spy", "run_id": "20260729-141230-a3f1",
  "status": "ok",          # ok | failed | timeout | skipped
  "exit_code": 0, "duration_s": 12.4,
  "started_at": "...", "finished_at": "...",
  "argv": ["..."],          # what actually ran, post package_command
  "kind": "python-module", "workdir": "...",
  "forced": false,          # true if --profiler overrode PACKAGE_PROFILERS
  "flamegraph": "profile.svg", # null unless declared AND the file exists
  "artifacts": ["profile.svg", "stdout.log", "stderr.log"],
  "host": "...", "git_sha": "..."
}
```

git_sha is best-effort: if git is missing or this is not a repository, the field is *omitted* rather than the run failing. flamegraph is null unless the profiler declares PROFILER_FLAMEGRAPH=1 *and* the file is actually on disk - declaring one is not the same as producing one.

summary.json

One file per invocation holding the arguments, per-status counts and the array of run records - so a downstream regression gate reads one file instead of globbing a tree.

A bug worth recording here

Re-running a *pinned* RUN_ID used to mix artifacts across attempts. Re-running build-7 with flamegraphs disabled produced a fresh profile.json beside the previous attempt's profile.svg, and meta.json claimed both. The run directory is now cleared first, guarded by a containment check against the output root, and the clearing is announced rather than silent.

SECTION 12

Retention

retention.py - deleting things carefully

--remove-output is a terminal action: prune, print what went, exit. --keep N keeps N runs **per (package, profiler) pair**, not N in total - so --keep 3 on a package with four profilers leaves twelve directories. That is what you want when comparing a profiler's history against itself.

This module deletes files, so it is the most defensive code in the tree:

Guard	Behaviour
Output directory sanity	Refuses if PROFILING_OUTPUT_DIR is unset, empty, relative, the filesystem root, or suspiciously shallow
Containment re-check	Every candidate is re-verified to resolve inside the output root immediately before deletion
Unknown directory names	Directories whose names do not match the run-id pattern are left untouched , never guessed at
Dry run	--dry-run lists exactly what would go and deletes nothing
Symlink repair	latest is re-pointed at the newest survivor, or removed when none remains

The consequence of the third guard is worth stating plainly: a run created with a custom RUN_ID opts *out* of automatic pruning, because the harness cannot tell it apart from a directory someone put there on purpose. Those must be deleted by hand.

The bug this module actually had

Survivors were computed as `runs[len(runs) - keep:]`. When keep exceeds the run count that index goes negative: --keep 5 against 3 runs sliced `runs[-2:]` and **deleted the oldest run it had been told to preserve**. It only misfires while `count < keep < 2x count`, so it would have looked like random data loss. The fix is `runs[-keep:]`, which clamps.

SECTION 13

Exit codes

One number that tells CI what happened

Code	Meaning
0	All runs succeeded. Skips do not affect this.
1	At least one workload failed or timed out
2	Usage error: unknown package or profiler, missing --profiler, bad flag, malformed .env
3	A required profiler binary is missing
4	A package init failed

The harness runs everything and then aggregates - it never stops at the first failure - and reports the **highest** applicable code. If one package's init fails (4) while another package's workload fails (1), the process exits 4.

Why skips are not failures

A skip means a profiler cannot handle a package's kind - asking py-spy to sample a non-Python executable, say. That is a statement about compatibility, not about the code under test. Making it a failure would mean - -package all --profiler all could never go green, which would make the most useful invocation the least usable one. Skips are recorded in summary.json with a reason, so nothing is hidden.

Why a malformed package is exit 2 and not exit 1

Exit 1 means "the code you are profiling has a problem". A package .env with a bad PACKAGE_KIND is a problem with the *harness configuration*, and a CI gate should treat those differently: one is a regression, the other is a broken pipeline.

SECTION 14

The four profilers

And the one that needed real work

Profiler	Kinds	Flamegraph	Notes
time	all three, incl. exec	no	GNU time -v: wall clock, CPU, peak RSS. Cheap enough to always run
py-spy	python	yes	Sampling. Needs ptrace. See below
viztracer	python	no	Deterministic timeline. Large artifacts, heavy overhead
line-profiler	python	no	Per-line timings via kernprof. Needs configuring

Flag spellings were verified against the installed tools - py-spy 0.4.2, viztracer 1.1.1, line_profiler 5.0.2 - rather than taken from documentation.

py-spy: the exit status problem

py-spy's exit status describes **py-spy**, not the program it ran, and the two are uncorrelated. Running one command ten times each:

```
child exits 0  -> py-spy exits  0 1 1 1 1 0 1 1 1 1
child exits 3  -> py-spy exits  1 0 1 1 1 1 1 0 0 1
```

The 1s on a *succeeding* child come from the flamegraph renderer reporting "No stack counts found" - a function of how long the workload ran, not of whether it worked. Since profiler_wrap's status is the workload's status, trusting py-spy's number would report broken workloads as successful and healthy ones as broken, at random.

So the profiler starts the workload itself and attaches py-spy to the resulting pid. The shell then owns the process and wait yields its exact code. The cost is that sampling begins a few milliseconds late.

The obvious alternative - keep launch mode and wrap the workload in a shell that records \$? - was measured and rejected: the extra fork makes py-spy miss the Python process entirely on **4 of 8** sub-second runs. Losing half the profiles is a far worse trade than a few milliseconds.

Interpreting a non-zero py-spy status

A non-zero status is still used, but only to tell its failure modes apart, and never by trusting the number:

What happened	How it is detected	Outcome
Attached, collected no samples	It still wrote an output file	Warning; the workload's status stands
Could not attach, workload still running	No output file, and the workload was alive when py-spy quit	Run fails - ptrace denied or similar
Could not attach, workload already gone	No output file, workload already finished	Warning; the workload's status stands

That last row is the attach-window race, and it is deliberately not a failure. A workload finishing in a few milliseconds occasionally beats the attach; failing on it would make fast packages flaky in CI. Distinguishing the last two rows is only possible while it happens, which is why the profiler waits on py-spy first and then checks the workload's state in /proc - a finished-but-unreaped child is still a signalable pid, so kill -0 cannot answer the question.

viztracer: artifact size

Stock defaults produced **106 MB** of JSON and 17 seconds of overhead from a 0.2-second workload. Capping the circular event buffer brings that to 22 MB and 2.6 seconds. The buffer is circular, so overflowing it costs the earliest events, not the run.

line-profiler: the one that is not zero-config

Without `LINE_PROFILER_TARGETS` (or `@profile` decorators) it produces an empty report. The harness warns - never fails - when a package selects it without configuring it. The warning is driven by a `PROFILER_WARN_IF_UNSET` declaration in the profiler's own `.env`, so the core knows nothing about line-profiler by name and the next tool with the same problem needs no core change.

SECTION 15

Is this too much code?

An honest accounting

2349 lines total, 1854 excluding comments and blanks. That is a fair thing to challenge. Here is where it actually goes.

Module	Lines	What it owns
<code>cli.py</code>	720	Parser (70), three terminal commands (116), <code>_run_pair</code> (113), <code>cmd_run</code> (71), dry-run printing, helpers, signals
<code>runner.py</code>	583	Two bash harnesses as string constants (~70), <code>execute</code> (96), target building (70), dry-run resolution (57), process-group control (60)
<code>discovery.py</code>	236	Scanning (40), typed accessors (110), selection (35)
<code>report.py</code>	228	<code>meta.json</code> (60), listings (40), table renderer (25), summary (40)
<code>initstate.py</code>	207	Fingerprint (40), decision table (30), execution (60)
<code>retention.py</code>	175	Safety checks (30), planning (50), applying (40)
<code>envfile.py</code>	200	<code>load_layers</code> entry point, sourcing (50), parsing (40), precedence (25)

Roughly a third is features, not core

Retention (135), init fingerprinting (163), dry-run resolution (~120), JSON listings and table rendering (~90), summary and meta writing (~90). That is about 600 lines - a third of the total - and none of it is optional if the tool is to have those features. The actual "run a thing under a profiler" path is closer to 400 lines: discover, layer the environment, build the argv, spawn bash, record the result.

About 70 lines are bug fixes

Signal handling, the SIGTERM-to-SIGKILL group escalation, and the run-directory reset. Not elegant, but each closes a leak that was demonstrated rather than imagined.

What was redundant, and is now gone

Two things this document previously listed as slop have been removed. `discovery._load_env` and `cli.load_global_env` were the same function differing only in which exception they raised - and both exceptions were caught on one line and produced the same exit code. They are now one `envfile.load_layers`.

And `_print_dry_run` used to call two resolvers backed by two nearly identical bash scripts, spawning two subprocesses for one printed line. They are now one `resolve_dry_run` and one subprocess.

The bigger change is in `cmd_run`: its inner loop body - the eight steps for one (package, profiler) pair - is now `_run_pair`, and `cmd_run` fell from 174 lines to 71. See Section 7.

Where the length is defensible

`cmd_run`'s 174 lines look like the problem and are not, for the reason given in Section 7: the ordering of its steps is the fragile part, and ordering is only visible when the steps are adjacent.

`envfile.py`'s 132 lines look like a lot for "source a file", but the precedence rule and the error attribution are the subtle parts, and both were arrived at by fixing observed problems.

SECTION 16

Testing

111 checks across five suites

Suite	Checks	Covers
acceptance	22	The specification's own acceptance list
bughunt	21	Quoting torture, hook ordering and failure, package_command, PACKAGE_TIMEOUT=0, malformed .env, unknown profiler, a real directory named 'latest'
bughunt2	16	All four environment layers individually, exit-code aggregation, 50k-line tee integrity by checksum, unicode, signal cleanup by process group
compliance	33	File tree, stdlib-only imports, every contract variable reaching hooks, cwd, hook ordering, init-once, meta.json fields, run-id format, run ordering
compliance2	19	Every config.env override, retention safety refusals, terminal-action behaviour, both py-spy failure classifications

Run twice end to end with no flakes. Two habits proved worth keeping:

Verify the test before believing the failure

Three early "failures" were the tests being wrong, not the code - a `grep -c` counting two lines instead of one, and a glob traversing the `latest` symlink so two runs looked like four. Each was checked before anything was changed.

Beware of measurement artifacts

`pgrep` and `ps | grep` match the measuring shell's own command line. This produced a phantom "py-spy leaks 2 processes" reading. The trustworthy numbers came from probing by process-group id instead.

SECTION 17

Deviations and judgement calls

Where the implementation departs from the specification, and why

1. py-spy uses attach mode, not the spec's literal command

The specification shows `py-spy record -- "${TARGET_ARGV[@]}"`. That form makes the workload's exit status unobservable, as Section 14 shows with measurements. The specification's own instruction to verify flag spellings against installed versions and adjust reads as licence for this, but it is a departure from the literal text and is documented in both the profiler and the README.

2. PATH-like variables are the one exception to layer-4 precedence

Strict precedence would make `PYTHONPATH="$MY_SRC:$PYTHONPATH"` in a package a no-op, contradicting the specification's own example. Section 5 gives the rule; it is called out in the README.

3. --dry-run exactness needs the profiler's cooperation

Achieved through an optional `profiler_dry_run` hook rather than by guessing. All shipped profilers implement it; third-party ones degrade to a labelled fallback instead of printing something that might be wrong.

4. viztracer gains a tracer_entries cap

Beyond the specification's `VIZTRACER_MAX_DEPTH`, because depth alone left 106 MB artifacts. The specification asks for conservative defaults and permits adjusting knobs, so this is within its intent.

Open item

The redundant dry-run harness described in Section 15 is known and unfixed. It is a contained change - delete `_RESOLVE_HARNESS` and `resolve_argv`, have `dry_run_command` return both values in one pass - and would want the 111 checks re-run afterwards.

The through-line

Two ideas carry this design. **Environment layering** means a package can tune a profiler for itself without either knowing about the other. **The target contract** means a profiler can be a prefix wrapper or an interpreter replacement without the core knowing which. Everything else - discovery, retention, init caching, exit codes - is bookkeeping around those two. If you understand Sections 5 and 6, the rest reads itself.